

EXECUTIVE MILESTONE REPORT: FG182 SWARM GOVERNANCE ACTIVATION

Formal verification & architectural certification of tri-agent consensus integration into the 18-Stage Execution Engine.

EXECUTIVE HEALTH INDEX	TRI-AGENT CONSENSUS	PIPELINE LATENCY	PIPELINE STAGES
98.00 / 100	100.0%	2.698 ms	18 / 18 PASS

1. Executive Summary

The **FG182 Multi-Agent Swarm Governance Kernel** has been successfully integrated into the 18-stage autonomous engineering runtime environment. During zero-trust verification testing, the kernel executed a complete pipeline pass in **2.698 ms**, achieving an Executive Health Index of **98.00/100.00**. Tri-agent consensus reached 100.0% alignment between the System Architect, Security Sentinel, and Compliance Auditor.

2. Cryptographic Governance & Verification Context

Execution ID:	KEXEC-7779FE0A
Institutional Rule ID:	IK-RULE-6560A0
Merkle Proof Hash:	d3bd90f69f6e52e401e418aeb6670f40e289b2d29fb87f9efbda1b424b5041ea
Governance Status:	GOLD PRODUCTION READY

3. Runtime Stage-by-Stage Audit Stream

Stage	Stage Name	Latency	Status	Action Details
[01]	Execution Request	0.001 ms	VERIFIED	Initial dispatch captured into pipeline boundary
[02]	Execution Context	0.001 ms	VERIFIED	Environment, authorization, and context isolation locked
[03]	Governance (FG182)	0.063 ms	VERIFIED	Tri-Agent Consensus CLEARED (Score: 100.0, Merkle proof generated)
[04]	Execution Plan	0.001 ms	VERIFIED	Decomposed task into 4 executable AST nodes
[05]	Scheduler	0.001 ms	VERIFIED	Queue positioning, time-slot, and priority allocation locked
[06]	Registry	0.000 ms	VERIFIED	Dynamic worker allocation and sandbox memory locks engaged
[07]	Workers	2.538 ms	VERIFIED	Parallel task compilation, build, and test execution completed
[08]	Event Bus	0.003 ms	VERIFIED	Real-time asynchronous telemetry event broadcast to cluster
[09]	Memory	0.009 ms	VERIFIED	Transactional state persistence written to Kernel Memory Store
[10]	Replay	0.003 ms	VERIFIED	Deterministic execution hashing & state match validated
[11]	Prediction	0.001 ms	VERIFIED	Defect probability (<0.001%) & latency variance projected
[12]	Learning	0.003 ms	VERIFIED	Institutional Learning Engine integration and sync complete
[13]	Optimization	0.001 ms	VERIFIED	JIT cache tuning and memory pool allocation optimized

[14]	Artifact Bus	0.004 ms	VERIFIED	Manifest publication and binary asset output verified
[15]	Reports	0.001 ms	VERIFIED	Compliance and technical execution audit report generated
[16]	Dashboard	0.001 ms	VERIFIED	Real-time UI metrics state stream synchronized
[17]	Exec Intelligence	0.005 ms	VERIFIED	9-Metric C-Suite synthesis generated (EOS Index: 98.00)
[18]	Institutional Knowledge	0.006 ms	VERIFIED	Permanent pattern store IK-RULE-6560A0 written to Swarm Memory

Prepared & Certified By:

Wilson Khanyezi
Founder & Chief Architect, Wilsy (Pty) Ltd

System Authorization:

Wilsy OS Swarm Governance Kernel
Status: 100% Cryptographically Verified